

CHOOSING BIG TECHNOLOGIES EXAMPLES FROM THE U.S. SPACE PROGRAM

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Three times in the evolution of the United States civilian space program – when President John F. Kennedy announced the decision to send Americans to the moon before a joint session of Congress on May 25, 1961, when the White House on January 5, 1972 announced that President Richard Nixon had approved the development of the space shuttle, a partially reusable space transportation system, and when President Ronald Reagan in his January 25, 1984 State of the Union address announced that he was directing the National Aeronautics and Space Administration (NASA) to develop a permanently manned space station within a decade – the United States has initiated a multiyear, multibillion-dollar program to develop a new space capability. The decisions to initiate these three programs – Apollo, shuttle, and space station – provide historical insight into the varying circumstances under which, at least in the United States and at least with respect to space, major technological undertakings are begun. The purpose of this paper is to carry out a comparative analysis of these three decisions.¹

POLICY CHOICE IN THE UNITED STATES

Leaders perform two different but interrelated tasks in selecting, from among competing alternatives, a particular course of action, a policy, for government to pursue. They must decide the following:

- 1) Is a proposed course of action a useful contribution to national goals and objectives, both in itself and in comparison with other possible actions?
- 2) Is that course of action politically feasible? Will it gain the support required for it to be acceptable to other powerful individuals and groups?

Several points about the character of the American system of policy choice need to be made explicit. They include:

- 1) The “national interest” is not a free-standing, easily identifiable, concept which exists independent of a particular situation and of particular individuals and groups. Rather, the national interest is defined through a process in which both the

long-term purposes of society and the shorter term interests of specific segments of that society are considered in the political competition for the commitment of resources. The national interest is not defined once and for all; it is constantly redefined, as external conditions, political leaders, and the influence of various interest groups change.

- 2) Among the various factors considered in national decision-making, the external situation (both external to the government and external to the United States) is in most cases the most influential. The actions of adversaries and allies, the condition of the economy, and the societal climate at a particular time all act to limit or to expand the range of feasible alternatives available to decision-makers.
- 3) The scope of the commitments required for large-scale enterprises, in both financial and time terms, requires agreement and cooperation among diverse segments of the political system. The pluralistic nature of American government means that a variety of individuals, interests, and organizations must cooperate over time for any large-scale undertaking to be approved and implemented. As the former NASA Administrator James E. Webb has noted:

In our pluralistic society and major public undertaking requires for success a working consensus among diverse individuals, groups, and interests. A decision to do a large, complex job cannot simply be reached "at the top" and then carried through. Only through an intricate process can a major undertaking be gotten underway, and only through a continuation of that process can it be kept going.²

How is such a "working consensus" reached in the American system? One way, the way which most closely approximates a conventional notion of rational decision-making, is for the leaders of government to agree on a definition of the national interest and national needs in a particular situation, and, based on that definition, compare various ways of pursuing the national interest and meeting national needs. An agency like NASA and its programs would be one means for achieving predefined national objectives, and NASA's task would be the relatively straightforward one of providing national leaders with the best quality space program or project to meet those objectives, given the funds available.

This "top-down" consensus is the exception, rather than the rule, in American policymaking. It seems to have been behind the approval of Project Apollo; President Kennedy for reasons of broad national strategy decided that the country should enter the space race, and asked NASA to tell him how to win. But when NASA waited for a political consensus to emerge on what the country should do in space after Apollo, it found that it was waiting for a train that was not likely to arrive at its particular station.

More often, what government agrees to do is the result of a complex, multichannel, time-consuming process of "bottom-up" consensus development, where the course of action chosen is the result of adjustments among conflicting perspectives and interests, each associated with one or more participants in the often confusing process of pluralistic policymaking. The genius of the American system of policy choice, according to many of its observers,³ is that this process of mutual adjustment, bargaining, and coalition-building both produces policy outcomes that are, for the most part, in society's interests, and at the same time minimizes destructive conflict among the many deeply felt points of view which are present in American society. The process by which the space shuttle was selected as NASA's major post-Apollo project closely

resembles this "bottom-up" model; NASA did not receive the shuttle assignment as a Presidential mandate, but rather only after three years of intense bargaining and compromising among the members of the space subgovernment.

The decision to begin the space station program combined some of the features of both the rational and pluralistic models of decision making. The proposal that the station be developed came from NASA's leaders; they spent most of two years in a wide ranging effort to develop a coalition in support of the station proposal that was broad and strong enough to provide a foundation for program approval. They were unable to do so. Two key potential constituencies, the Department of Defense and the scientific community, made it clear that they did not support the station, and the leaders of those two constituencies actively opposed the station program. However, when the decision of whether or not to approve the station was brought before the President, he chose to proceed with the program, primarily because it fit his own image of what was in the national interest. Moreover, he embraced it as a personal initiative, adding the power and prestige of the Presidency to the weak bureaucratic coalition supporting the station program.

The following is a more detailed description of the ways in which the commitment to Apollo, the space shuttle, and the space station were made. The contrasts in the decision-making process in these three cases are stark, and are intended to suggest the variety of ways in which a decision to commit substantial public resources to a large-scale enterprise in space can occur.

APOLLO AS A RATIONAL CHOICE

On occasion, it seems possible to understand government decision-making as similar to the behavior of a rational individual seeking the best means to achieve his goals. The government is seen as a unified actor that chooses its policies according to a procedure involving the following steps:

- 1) A problem, opportunity, or need is perceived and defined.
- 2) Related goals and objectives are specified and ranked in order of their relative importance.
- 3) Alternative policies for achieving those ends are identified.
- 4) The consequences, both favourable (benefits) and unfavorable (costs), that follow from each policy alternative are calculated.
- 5) The alternative with the best mixture of costs and benefits over the whole range of goals involved is chosen as government policy.

Under most conditions, the checks and balances built into the U.S. political system provide barriers to the real-world operation of this procedure. Too many interests must be consulted for a single set of goals to be identified and ranked, and for alternatives for achieving them to be compared objectively. However, in times of crisis (when situations arise without the benefit of extensive prior planning of an appropriate response, which allow only a short time for response, and where the issues at stake are of great importance), many of the barriers to rational, "top-down" decision-making disappear.

Such a situation occurred in April 1961. The self-image and international standing of the United States had been stung in rapid succession by the success of the Soviets in orbiting the first human, and by the failure of the United States to follow through in support of the invasion of Cuba at the Bay of Pigs by U.S.-trained forces. In a memorandum dated April 20, 1961, President John F. Kennedy defined his objective:

Do we have a chance of beating the Soviets by putting a laboratory into space, or by a trip around the moon or by a rocket to land on the moon, or by a rocket to go to the moon and back with a man? Is there any other space program which promises dramatic results which we could win?⁵

What Kennedy sought was to demonstrate to the world that the United States remained the leading nation in technological and social vitality. Almost equally important, though not as clearly articulated, Kennedy saw Apollo as a means of restoring American pride and self-confidence. The Soviet Union's surprising demonstration of technological and strategic strength through its series of space firsts leading to Yuri Gagarin's flight had shaken the self-image of the United States as the unchallenged technological leader of the world. According to close Kennedy associate Ted Sorensen, the Gagarin flight and the reaction to it "demonstrated to the President the importance of going ahead with an all-out space effort and the willingness of the country and the Congress to back such an effort."⁶ The President told a White House meeting called two days after the Soviet success (and before the Bay of Pigs) to discuss ways of the United States catching up that "there's nothing more important."⁷

Kennedy found unacceptable the possibility that the United States would take second place to the Soviet Union in a critical area of human activity. Kennedy's science advisor Jerome Weisner stressed this fact:

If Kennedy could have opted out of a big space program without hurting the country in his judgment he would have. Maybe a different kind of man could have said to the country, "Look, we are going at our own pace. We are going to let the Russians be first. We don't care." But Kennedy said, "If we could afford to do something else, we would do it. If we can't we had better get back where we belong." I think he became convinced that space was the symbol of the twentieth century. It was a decision he made cold-bloodedly. He thought it was right for the country.⁸

After two weeks of assessing alternative answers to these questions, Kennedy's advisors, led by Vice President Lyndon Johnson, agreed that the United States had at least a 50-50 chance of winning a competition to complete a successful manned lunar expedition, and that no other alternative provided a better combination of achievement, risk, and cost. Kennedy accepted this assessment, and the Apollo program was born.

The memorandum to the President recommending the lunar landing effort was signed by Johnson, NASA Administrator James Webb, and Secretary of Defense Robert McNamara. It stated clearly the rationale underpinning the enterprise:

It is man, not merely machines, in space that captures the imagination of the world All large-scale projects require the mobilization of resources on a national scale. They require the development and successful application of the most advanced technologies Dramatic achievements in space, therefore, symbolize the technological power and organizing capacity of a nation. It is for reasons such as these

that major achievements in space contribute to national prestige.⁹

Space achievements developed prestige, they asserted, in the power struggle between the United States and the Soviet Union; the United States should thus undertake a manned mission to the moon, even if the scientific or military grounds were lacking.

Major success, such as orbiting a man as the Soviets have just done, lend national prestige even though the scientific, commercial, or military value of the undertaking may be marginal or economically unjustified Our attainments are a major element in the international competition between the Soviet system and our own. The non-military, non-commercial, non-scientific but "civilian" projects such as lunar and planetary exploration are, in this sense, part of the battle along the fluid front of the cold war.¹⁰

Kennedy accepted these arguments. In announcing his decision on May 25, 1961, he told the Congress and the nation:

If we are to win the battle that is going on around the world between freedom and tyranny, if we are to win the battle for men's minds, the dramatic achievements in space which occurred in recent weeks should have made clear to us all, as did the Sputnik in 1957, the impact of this adventure on the minds of men everywhere who are attempting to make a determination of which road they should take We go into space because whatever mankind must undertake, free men must fully share.¹¹

Apollo was, of course, a total success in achieving its objective of landing Americans on the moon before 1970; it has also been selected as the top engineering achievement of the twentieth century by the U.S. National Academy of Engineering. But the link between space accomplishment and high priority national interests proved fragile. The decision to initiate the Apollo program had little to do with the benefits of a long-term commitment to space development and exploration; it was driven by considerations external to the space program. Once those conditions had changed (in no small part due to the success of Apollo and its predecessor programs), the political base for continuing the kind of mobilization of resources that produced Apollo proved impossible to sustain. The Apollo decision is best understood as national policy, not space or technology policy; the space program was used as a *means* to broader national purposes, not supported on its own merits. Thirty years after the decision to begin Apollo, the debate over those merits still continues.

PLURALISTIC POLICYMAKING AND THE SPACE SHUTTLE¹²

The design of the U.S. government makes the kind of top level goal-setting and means-comparing implied by the rational model almost impossible under "normal operating conditions." The U.S. system puts a higher value on avoiding concentrated power than it does on efficiency or effectiveness; the U.S. government is not intended to operate according to rational principles, at least as they are understood by scientists and engineers.

The "normal" process of policymaking involves a wide variety of participants. It is characterized by bargaining among players positioned within various government

organizations. Individuals and groups outside government participate in this process and can be very influential, but their power lies primarily in influencing those within government who control the resources required to undertake a new course of action. Decisions are almost always made not by one central decision-maker, but by a process of interaction among various government organizations and individual political actors.

The January 1972 decision by President Richard Nixon to develop the space shuttle is an excellent example of the normal policymaking process in operation, just as Apollo can be understood as the result of a rational choice process. In the shuttle decision, major participants were:

- 1) NASA, both as an engineering organization eager to take on a new challenging technology development job after Apollo and as a government agency interested in maintaining its budget, institutional base, and status.
- 2) Some in the Department of Defense, attracted by the potential of the proposed shuttle for various national security missions in space.
- 3) The aerospace industry, interested in undertaking another major effort along the lines the Apollo.
- 4) The Congress, still supportive of space but unwilling to approve another Apollo-like project aimed at, for example, human planetary exploration.
- 5) Scientists skeptical of the value of or need for another manned program to follow Apollo.
- 6) Analysts who, for the first time, examined a major space initiative in terms of its cost effectiveness.
- 7) The Office of Management and Budget, protective of the budget and unconvinced that the shuttle was a good investment of public funds.
- 8) The President's Science Advisor and his supporting staff and advisory committees, who believed that some sort of shuttle program was an appropriate post-Apollo space initiative, but who were skeptical of the NASA-defined shuttle as being the best approach to lowering the cost and increasing the ease of access of space.
- 9) President Nixon and his policy advisors, skeptical of the future political payoffs from major space programs and wanting to reduce the size of the NASA budget from its Apollo levels, but unwilling to take the United States out of manned spaceflight, and eager for the political payoffs from the employment impacts of programs such as the shuttle in key electoral areas such as California.

The shuttle decision was the end product of a high-pressure, broadly-based, sometimes confused debate that extended from early 1969 to early 1972, reaching a peak in the second half of 1971. The shuttle which President Nixon finally approved for development was dramatically different in both design and estimated cost from that which NASA had originally hoped to develop. The final design emerged from a process of negotiation, compromise, and conflict. It had the rationale, technical characteristics, and cost implications required to gain the support of the President and his advisers, the Department of Defense, and a majority of Congress, while still meeting most of the needs of NASA and its contractors. This coalition was able to

overcome continuing opposition from the scientific community and the President's budget office, and thus provided *just* enough support of the program to gain Presidential approval.

The shuttle was approved at the culmination of the following events:

- 1) The shuttle program first appeared as a major part of NASA's plans in 1969. At that time, it was seen almost exclusively in terms of providing low-cost transportation between the surface of the earth and space stations in near-earth orbit. Such low-cost transportation was thought to be necessary if long-term human occupancy of space was to be affordable, but developing space stations was the goal and the shuttle only a means to it.
- 2) In mid-1969, NASA identified a manned Mars mission in the 1980s as a central theme of its post-Apollo planning; this represented a shift away from portraying a space station and larger space base as the major post-Apollo goals. There were several reasons for switching to the Mars emphasis. NASA had earlier tried to gain support for a large space station program on the basis of its being the next logical step in human space flight. Neither that rationale nor any other that NASA had tried seemed to get much support. By picturing a space station as a necessary precursor to Mars missions in the eighties, NASA Administrator Thomas Paine hoped to provide a convincing rationale for the station's urgency. Not only the space station, but also the space shuttle, the development of nuclear rocket engines, and the retention of the large Saturn V as a booster were required if an early Mars landing were to be approved as a national goal. However, the policymaking system said a resounding "No" to NASA's Mars proposals; NASA found out just how limited the support was for major new space initiatives in the White House, the Congress, and other segments of the U.S. society, and the next several years were spent trying to adjust to this reality.
- 3) The shuttle design that emerged from preliminary studies featured two components, both reusable and both operated by human crews. One was a booster roughly the size of a 747 aircraft, which would provide the initial thrust to lift the shuttle off the earth's surface. After launch, the booster stage would be flown by its crew to a landing near the launch site and refurbished for its next flight. The second stage was an orbiter of 707 aircraft size, which would be carried atop the booster and which, after separation from it, would use its own engines to lift it into orbit. Once in space, the orbiter would rendezvous with a space station to transfer supplies or people and then return to a landing on earth.
- 4) During 1970 it became clear to all concerned that NASA would not get approval to develop simultaneously both a space station and the shuttle, due both to the high cost of the two programs and to a sense in Congress and the White House that the space program no longer had sufficient priority to justify such costs. Although NASA's preference seems to have been to proceed with space station development, its leadership realized that the space shuttle was the program more likely to get Presidential approval. Thus, NASA deferred its space station program indefinitely and decided to concentrate on gaining approval for developing a two-stage reusable shuttle as its major program of the 1970s.
- 5) After dropping its space station plans, NASA had to develop a rationale stressing the shuttle's utility, in comparison with other potential means, for performing a

variety of roles related to automated space missions for NASA, the Department of Defense, intelligence agencies, other countries, and commercial users. In addition, the shuttle could act in some ways as a substitute for a more permanent space station. In justifying the shuttle, NASA and its contractors prepared an elaborate economic analysis. This was the first time such an analysis had been used as a tool in gaining support for a major space effort.

- 6) NASA's plans to develop the two-stage reusable shuttle were dealt a fatal blow when the Office of Management and Budget indicated in mid-1971 that NASA's budget would have to remain constant for at least the duration of the Nixon administration. This made it impossible to plan a program that would require over \$2 billion annually at its peak, and it set off a flurry of studies and re-evaluations aimed at identifying a shuttle design offering the maximum capability for the minimum overall and annual development costs, even if it cost more to operate than other alternatives. Such a design still had to be preferable to alternative systems for accomplishing the space missions of the 1980s. It was during this process, extending through the second half of 1971, that, according to one participant, "everybody was a shuttle designer." There were rapid changes in cost estimates and design characteristics, and disagreements over the shuttle design extended for several months after the Presidential announcement of the decision to proceed.
- 7) The decision to approve the shuttle was made by the President and his policy advisers, despite the continued skepticism of the White House budgetary and technical staff, on the grounds that the United States should develop the new capabilities required to keep it in a leadership position in space during the 1980s, and that any United States space program should include a manned component. In the final decision, concerns of cost effectiveness played a relatively minor role.

As a result of the debate preceding this decision, the shuttle program that was finally approved had substantially lower technological risks and development costs than the program NASA had been proposing only six months earlier. The final configuration did not sacrifice any of the new capabilities for operating in space offered by the original configuration. Indeed, maintaining parameters such as payload bay size, lifting capacity, and cross-range capability was essential if the Department of Defense was to continue to support the shuttle program. But NASA had been forced by continuing pressure from the White House budget office to bring forward a shuttle design that stressed low development cost; the Nixon administration was willing to accept higher operating costs, to be paid by future users of the shuttle, with a different President in the White House, as long as the investment made in the system while it was in office was kept to a minimum.

The space shuttle was approved as a means for operating in space without any extensive debate over what the goals of space operations in the 1980s might be. This is typical of normal policymaking in the American system, where debate over means often acts as a surrogate for controversy over ends. Much attention was given to the details of various shuttle configurations, and certainly participants in the shuttle debate brought to it various images of the likely future in space, but these images were never made explicit and then subjected to the same searching analysis as was the shuttle itself. It is the nature of the American system of government that a hierarchical ordering of goals and priorities very seldom occurs. What substitutes for such an

ordering is the complicated process of cooperation and competition among policymakers that took place in the case of the shuttle, in which a "strain toward agreement" overrides conflict about preferred goals.

SPACE STATION - A WEAK COMMITMENT

When a new set of managers, Administrator James Beggs and Deputy Administrator Hans Mark, assumed control of NASA in 1981 as the government of Ronald Reagan came to Washington, they recognized that the space shuttle development program was nearly at an end, and that a major new project was needed if NASA was to retain its identity as a developer of new space technology. As noted above, NASA had deferred its attempts to gain approval for a space station in order to be able to proceed with the shuttle; it was not surprising that, as the shuttle development program neared completion, the agency would resurrect the station proposal. Thus Beggs and Mark announced that gaining approval for the development of a permanently manned space station would be one of their highest priority objectives. Having made this announcement before they were even officially approved as NASA leaders, Beggs and Mark then had to design a strategy to secure that approval.

That strategy had two main paths. One followed the pluralist approach, attempting to engage in early planning of the station program all those interests who might want to participate and thereby gain their support when the White House and the Congress considered whether or not to approve it. Among those who became involved were:

- 1) NASA field centers who, as the agency's budget had grown smaller during the 1970s, had become intensely competitive with one another for the few new projects being initiated; one goal of Beggs and Mark was to plan a station program large enough to provide substantial work for several centers and thereby reduce internal conflicts within NASA.
- 2) Other government agencies, and particularly the Department of Defense, that might make use of the station and perhaps even share in the costs of its development.
- 3) The space science community, dependent on NASA for financial support and for access to space for its experiments, but skeptical of the scientific value of human presence and determined to avoid its experience with the shuttle program, where budget overruns had been funded at the expense of new science missions and where the requirement of launching their experiments abroad the shuttle had added cost and complexity.
- 4) Aerospace industry firms, who were given contracts by NASA to identify potential users and missions for a space station. Industry, in turn, solicited ideas for station use from a wide variety of organizations, including many not previously involved in space.
- 5) Potential international partners who, in a change from the past practice of not involving them until a U.S. decision to go ahead with a program had been taken, were invited to carry out parallel studies of station utilization and of hardware contributions they might make to a station program.

This pluralist strategy was complemented by an approach focussed on President Ronald Reagan and his policy advisers. NASA Administrator Beggs, in particular, believed that the President could be convinced to approve the station even if the "bottom up" process of consensus building failed to provide a broad basis of support for the program.

The consensus-building process went on throughout 1982 and most of 1983. Its outcome was clear; support for going ahead with the station was narrowly concentrated in NASA and its contractor allies and in the space agencies of potential international partners. Other federal agencies saw no need for an expensive orbital laboratory in terms of their future plans, and the leadership of the Department of Defense, including Secretary of Defense Caspar Weinberger, actively opposed the project. The scientific community overall was distinctly cool to the station proposal, even though NASA had created a science advisory group so that the views of the community could be taken into account. The attempt to discover unique or valuable missions for a station had centered on the speculative possibility of high economic returns from materials experiments and manufacturing activities in the very low gravity environment aboard the station; in 1983, predictions of a new multibillion dollar space manufacturing industry were being widely cited by those in NASA trying to sell the station program. The interagency decision process, managed by the National Security Council, that was supposed to bring to the President a recommendation on whether to approve the station, was stalemated by the determined opposition of the Department of Defense, the Office of Management and Budget, and the President's Science Adviser.

Fortunately, at least from NASA's point of view, its second strategic approach, gaining direct Presidential approval for the station program, bore fruit. NASA had been seeking such approval since early 1982, and had hoped to obtain it as the President met a returning shuttle mission and announced his new space policy at Edwards Air Force Base on July 4, 1982. The President and his advisers judged that such approval was premature, however, and at the welcoming ceremony the President said only that the United States had to establish "a more permanent presence in space;" he did not say that such presence had to involve humans or specifically mention a space station.

In April 1983 President Reagan, after a NASA briefing on the reasons for going ahead with a space station, personally signed a directive setting out the terms of reference for the National Security Council study mentioned above; his signature was taken as an indication of his particular interest in the station program. In August 1983, NASA's allies within the White House were able to schedule a luncheon meeting between the President and a group of top aerospace industry officials, who stressed their belief that the station was a needed program and that the concept of space manufacturing using the station held significant economic promise. In December, the same internal allies found a way around the National Security Council stalemate on the shuttle; they used a different bureaucratic path, a meeting of the Cabinet Committee on Commerce and Trade, to get the station proposal before the President. In this December 1 meeting, the President heard a carefully prepared NASA briefing on the station that emphasized its role in preserving American space leadership, viewed an elaborate model of the facility, and was exposed to the views of several of his advisors that going ahead with the station made good sense. Opponents of the program also had their say, but in a muted way.

The president did not announce his decision at the meeting, but within a few days NASA Administrator Beggs learned that Reagan had decided to go ahead with the station program. In the following six weeks, the President decided not only to approve the program, but to embrace it as a personal initiative, particularly in the context of inviting international participation. By the time the President went before a joint session of Congress for a nationally-televised State of the Union address on January 25, 1984 and announced that "tonight I am directing NASA to develop a permanently manned space station and to do it within a decade," he had already sent messages to Heads of States around the world informing them of his decision and his intent to invite their country's participation in the program. In March 1984, NASA Administrator Beggs made hurried trips to Europe, Japan, and Canada to extend the President's invitation in person, and Reagan reiterated his personal interest in the undertaking at the June 1984 Economic Summit meeting in London.

In this way, the space station became not only the successor to Apollo and the space shuttle as the centerpiece of the U.S. space program but also the focus of the largest experiment in international technological cooperation ever undertaken. But the uncertain foundation upon which the program had been planned did not disappear with Ronald Reagan's endorsement. In 1961, John Kennedy decided to go to the moon only after he was assured that there was a sufficient base of political support to sustain the program through to completion; more than two decades later, Ronald Reagan decided to press forward with the space station in spite of strong opposition within his administration to going ahead. In this way, there were strong similarities between the station and shuttle decisions; a main difference was that Ronald Reagan was personally much more identified with the space station than Richard Nixon had been with the space shuttle.

Howard McCurdy has asked whether the decision to build the space station was "too weak a commitment."¹⁴ He suggests that, given the lack of strong national and Presidential support for major space initiatives since the end of the Apollo program (and even during its final years),

NASA officials have learned to accept programme approval under conditions which make long-term commitments difficult to achieve. This has created a significant dilemma for advocates of major new space initiatives—whether to pursue approval for programmes such as the Space Station under conditions which erode the long-term stability necessary to complete them, or not to pursue the programmes at all¹⁵

This suggestion neatly encapsulates the primary lesson of the space station (and perhaps the space shuttle) decision; that *both* an adequate political base among those interests and organization involved in financing, approving, developing, and using a major space capability *and* the active support of the President (not just passive approval, as was the case with respect to the space shuttle) are necessary elements for success. Obtaining both for any new space project in the years since John Kennedy sent Americans to the moon three decades ago has proven impossible, and the current outlook with respect to President George Bush's 1989 proposal that the United States undertake a long term program of human exploration of the solar system is no more promising.

Apollo was born in the midst of a perceived crisis; going to the moon was an acceptable answer to President Kennedy's search for a way out of that crisis. But no crisis

since 1961 has been such that a space program has been an appropriate response, and in non-crisis situations the United States still appears uncertain about the priority it wishes to assign to space exploration. The outlook in the United States for starting major new space programs in coming decades is clouded.

Notes

- 1 This paper is a revision and updating to include a discussion of the space station decision of John M. Logsdon, "The Apollo and Space Shuttle Decisions," in Theodore R. Simpson, ed., *The Space Station: An Idea Whose Time Has Come* (New York: IEEE Press, 1985).
- 2 James, E. Webb. *Space Age Management: the Large-Scale Approach* (New York: McGraw Hill, 1969), p. 89.
- 3 For a discussion of this perspective, see Charles Lindblom, *The Policymaking Process* (Englewood Cliffs, N.J.: Prentice-Hall, 2nd ed., 1978) and Charles Lindblom, *The Intelligence of Democracy*, (New York: The Free Press, 1965)..
- 4 This discussion is based primarily on John M. Logsdon, *The Decision to Go to the moon: Project Apollo and the National Interest* (Cambridge, Mass.: The MIT Press, 1970). All quotations in the following discussion of Apollo are taken from that book.
- 5 *Ibid*, pp. 109-110.
- 6 *Ibid*, p. 107.
- 7 *Ibid*, p. 106.
- 8 *Ibid*, p. 111.
- 9 *Ibid*, p. 125.
- 10 *Ibid*, p. 126.
- 11 *Ibid*, p. 128.
- 12 This analysis is based primarily on John M. Logsdon, "The Decision to Develop the Space Shuttle," *Space Policy*, Vol. 2, No. 2 (May 1986) and John M. Logsdon, "The Space Shuttle Program: a Policy Failure," *Science*, May 30, 1986.
- 13 This discussion, in addition to drawing from the author's observations, depends heavily on the excellent study by Howard E. McCurdy, *The Space Station Decision Incremental Politics and Technological Choice* (Baltimore: Johns Hopkins Press, 1990). 1 See also Hans Mark, *The Space Station: a Personal Journey* (Durham, N.C.: Duke University Press, 1987) for one participant's account of the station decision.
- 14 Howard E. McCurdy. "The Decision to Build the Space Station: Too Weak a Commitment?" *Space Policy*, Vol. 4, No. 4 (November 1988), pp. 297-306.
- 15 *Ibid*, p. 297.